Following the same calculation, we can obtain the entropy for this new distribution as

$$\sum_{i=2}^{100} 0.0101 \log 0.0101 = 6.63$$

Again, this is the posterior entropy of L in the second case where z is not present in the first location. The last step we need to take in order to compute the overall posterior entropy after inspecting the first location is to take the average of the two cases:

- If z is in the first slot, which is one percent likely, then the posterior entropy is 0.
- If z is not in the first slot, which is ninety nine percent likely, then the posterior entropy is 6.63.

Taking the average, we have $0.01\ (0) + 0.99\ (6.63) = 6.56$. So, on average, we expect to see a posterior entropy of 6.56 when we choose to look into the first element of the array. Now, in order to see whether looking at the first element is the optimal decision, or if there's a better location that will help us obtain more information about L, we need to repeat this procedure for the other locations in the list. Specifically, for a given location, we need to

1. Iterate over the different scenarios that could happen when we inspect the location,
2. Compute the posterior entropy of the distribution of L for each scenario,
3. Compute the average posterior entropy across the scenarios based on how likely each of them is.

Let's do this one more time for, say, the tenth location x_{10}; the corresponding diagram is shown in Figure 6.8. While things are slightly different from what we just went through, the underlying idea is still the same. First, there are various scenarios that can happen when we look at x_{10}:

1. The tenth element x_{10} can be greater than z, in which case we can rule out the last 91 elements in the list and focus our search on the first 9 elements. Here, each of the 9 locations has 11 percent of containing z, and the posterior entropy, by using the same formula, can be computed to be roughly 3.17.
2. The tenth element x_{10} can be exactly equal to z, in which case our posterior entropy is once again zero.
3. The tenth element x_{10} can be less than z, in which case we narrow our search to the last 90 elements. The posterior entropy in this case is around 6.49.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>